AI Engines – Input / Output Specification

- **What data it receives (Input)**
- **What processing it does**
- **What result it returns (Output)**

This is called I/O Specification.

1️⃣ ATS AI Engine (Resume Parser)

Input:

- **Resume file (PDF / DOC)**
- **Candidate ID**

Processing:

- **NLP text extraction**
- **Skills detection**
- **Education & experience identification**

Output:

- **Candidate name**
- **Skills list**
- **Education details**
- **Experience years**
- **Structured JSON profile**

2️⃣ Screening AI Engine (Job Matching)

Input:

- **Candidate skills**
- **Job requirements**
- **Experience level**

Processing:

- **Skill similarity algorithm**

- **AI ranking logic**

Output:

- **Match score (0–100)**
- **Candidate rank**
- **Job fit label (Good / Average / Poor)**

3️⃣ Interview Intelligence AI Engine

Input:

- **Interview video / audio**
- **Speech transcript**

Processing:

- **Speech-to-text**
- **Answer keyword evaluation**
- **Communication analysis**

Output:

- **Technical score**
- **Communication score**
- **Interview feedback summary**

4️⃣ Behaviour Analysis AI Engine

Input:

- **Facial expression frames**
- **Voice tone data**

Processing:

- **Emotion detection model**
- **Confidence estimation**

Output:

- **Confidence level**
- **Emotion state**
- **Personality indicators**

5️⃣ Decision & Scoring AI Engine

Input:

- **ATS structured data**
- **Screening score**
- **Interview score**
- **Behaviour score**

Processing:

- **Weighted scoring algorithm**
- **Final ranking logic**

Output:

- **Final candidate score**
- **Hire / Reject recommendation**
- **Priority ranking**

Why This I/O Specification Important

- **Helps backend developers know how to call AI APIs**
- **Helps AI team design clear model responsibilities**
- **Improves scalability & debugging**
- **Enables microservices architecture**